

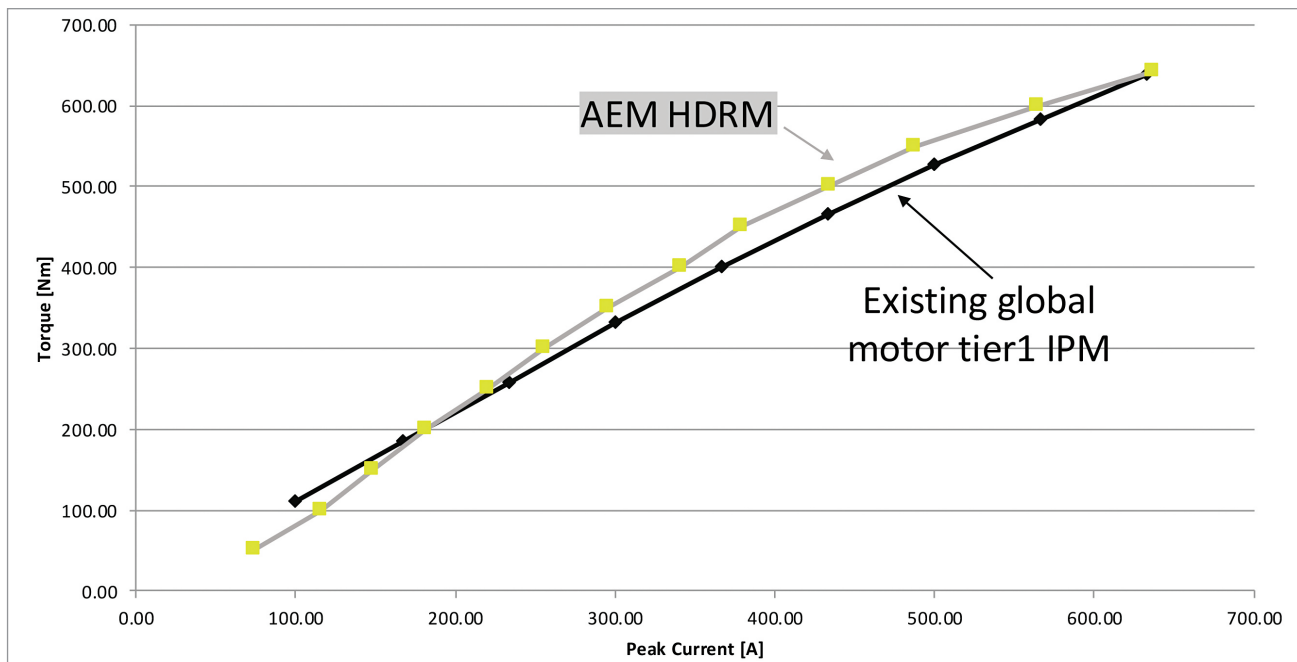


Fig. 3: HDRM and IPM comparison (Courtesy: AEM, Sterling GTake)

in India with high domestic value addition. Only two components are imported, and those can be sourced from multiple global suppliers.

On the subject of production readiness, Wadhwa stated, “We’ve not built assembly lines yet because platforms vary by vehicle type. But we are close to a decision on which direction to invest.” He emphasised that the technology is both scalable and versatile.

During the Q&A session, James addressed the relatively slow adoption of magnet-free motors. “This is still new tech. We began in 2017. Earlier versions had performance compromises, but our designs do not.” Asked to compare with permanent magnet synchronous motors (PMSMs), he noted, “Half the cost of high-volume motors comes from magnets and copper. Removing those gives us a clear cost advantage.”

Another question raised was whether India’s EV industry could realistically shift to this new motor type, especially amid continued supply chain challenges related

to China. Jaideep responded, “We believe this is a true drop-in replacement. We can directly replace an internal permanent magnet motor one-to-one.”

On the topic of recycling rare-earth elements, the experts were pragmatic. “Most of the recoverable rare-earths are actually from hard disk drives... So, while I am a great believer in trying to recycle and recover what we can, unfortunately, that alone is not going to solve the problem,” said James.

He noted another complication, “You certainly cannot recover all of the magnet material very easily,” particularly due to demagnetisation after prolonged use.

Concerns were also raised about Chinese dominance in the EV sector and whether this could block wider adoption of AEM’s technology. In response, James remarked, “My understanding is that none of them are making any money... So do not assume that just because they have access to rare-earths, they do not see the benefits of moving away

from them.”

Regarding early adopters in India, James stated that while specifics remained confidential, he would be “very, very surprised if we do not soon see this technology out there on the streets of India.”

Wadhwa shared similar views on adoption, noting that AEM had already found success with commercial vehicle manufacturers globally, and India could follow suit. However, he admitted the timeline remains fluid. “Earlier, companies had plans to introduce these motors in 2028 or 2030... What is different now is that people are asking, ‘What can you do now?’”

The session ended on a realistic but optimistic note. “Magnet-free motors have always been on the radar, but the urgency has changed.” **EFY**

Based on a white paper and webinar titled ‘Rare Earth Magnet-Free Motors: Next Generation Electric Traction,’ organised by Sterling GTake E-Mobility (SGEM) and Advanced Electric Machines (AEM) on 11 July 2025. Curated by Shubha Mitra, Assistant Editor at EFY.